B.Tech. (Computer Engineering) V Semester Examination, 2021

CEN-504 **Object Oriented Programming** **Roll No.……**

Paper No. CEN-504

Time: Three Hours

Maximum Marks: 60

Note: Attempt any two parts from each question.
Make suitable assumptions wherever necessary.

Q.1 CO1	**(a)**	Define the term *Object Oriented Programming* briefly. Write a program in C++ to handle the details of a cricket *player* using array of objects. The class must have following data members: *Player code, Player name, runs, innings- played and number of times not out.* Include member functions to perform following operations: ➢ Input details of players. ➢ Display details of all players. ➢ Display average runs of a single player. ➢ Display average runs of all players. (Use function overloading) The program must be able to handle the details of 10 players.	**6**
	(b)	Write an interactive program in C++ using self-referential classes. The program should support insertion at end and deletion in the beginning on a single link list.	**6**
	(c)	**(i)** Differentiate among private, public and protected members of a class in the context of Object Oriented Programming. Also give a suitable example illustrating accessibility / non-accessibility of each using a class and an object in C++ / Java. **(ii)** Explain briefly with the help of suitable code snippets: ➢ Inline functions ➢ Copy constructor ➢ Dynamic constructor	**6**
Q.2 CO2	**(a)**	Design and implement a class named *Date* that contains three private data members month, day and year that provides the following functionality - Creating date with default values (zero for day, month and year). - Setting value of date at the time of creation. - Changing values (day, month or year separately or combine) of any date. - Getting values (day, month or year separately) of any date. - Displaying date value. - Keep track of how many dates are created in a program and a function to access this value. - Overload the following operators for comparing two dates. One date is less than other (<) One date is greater than other (>) Two dates are equal (==) Two dates are not equal (!=) Increment 1 in day value of date (++) both pre and post Decrement 1 in day value of date (- -) both pre and post	**6**

	(b)	Create a class *Time* which contains: – hours – minutes – second Write a C++ program using operator overloading for the following: (i) = = to check whether two Times are same or not. (ii) >> To accept the time. (iii) << To display the time. Implement all relevant constructors also.	**6**
	(c)	Write a program to overload subscript operator such that out of bounds memory accesses can be taken care off. Why does an overload [ ] operator can return a reference or a non-reference argument both?	**6**
Q.3 CO3	**(a)**	In a bank, different customers have saving account. Some customers may have taken a loan from the bank. So bank always maintains information about bank depositors and borrowers. Design a base class *Customer* (name, phone-number). Derive a class Depositor (accno, balance) from Customer. Again derive a class Borrower (loan-no, loan-amt) from Depositor. Write necessary member functions to read and display the details of 'n' customers. Implement relevant constructors also.	**6**
	(b)	Differentiate between method overriding and method overloading with the help of suitable program in C++ /Java.	**6**
	(c)	Demonstrate the concept of virtual base class with the help of a suitable program.	**6**

Q.4 CO4	**(a)**	What is the only effect that the abstract keyword has on a class? What are the two main effects of declaring a method abstract?
		Use the following three class definitions to answer the remaining questions.
		//filename: Mammal.java
		public abstract class *Mammal*{
		public void breathe(){
		System.out.println("Mammal is breathing");
		}
		public abstract void nurse();
		}
		//filename: Cat.java
		public class *Cat* extends *Mammal*{
		public void breathe(){
		System.out.println("Cat is breathing");
		}
		public void nurse(int t){
		System.out.println("Cat is nursing");
		}
		}
		//filename: Whale.java
		public class *Whale* extends *Mammal*{
		public void breathe(){
			6

```
        System.out.println("Whale is breathing");
        super.breathe();
    }
    public void nurse(){
        System.out.println("Whale is nursing");
    }
}
```

Justify your answers properly.

(i) The Whale class compiles fine, but the Cat class does not. Why not? What can be done to fix the problem?

(ii) Is the following statement valid?

```
Mammal m = new Whale();
```

(iii) What is the output of the following statements?

```
Whale w = new Whale();
w.breathe();
w.nurse();
Mammal m = w;
m.breathe();
m.nurse();
```

(b) Show the use of following keywords of Java with example code snippets: **6**

(i) super
(ii) static
(iii) final

(c) Use the following class definitions to answer the next four questions. **6**

```
public class Vehicle{
    public int numOfWheels;
    public Vehicle(int x){
        numOfWheels = x;
    }
    public void drive(){
        System.out.println("Driving a vehicle");
    }
}
public class Car extends Vehicle{
    public int numOfDoors;
    public Car(int d, int w){
        numOfDoors = d;
        super(w);
    }
    public void drive(){
        System.out.println("Driving a car");
    }
}
```

		Justify your answers properly.
		(i) The Car class does not compile. Why? How can it be corrected?
		
		(ii) (Assume from here on that the problem with the Car class has been corrected.)
		What is the output of the following statements?
		Car audi = new Car();
		Car.drive();
		
		(iii) What is the output of the following statements?
		Car porsche = new Car(2, 4);
		porsche.drive();
		
		(iv) What is the output of the following statements?
		Vehicle v = new Vehicle(18);
		v.drive();
		v.numOfDoors = 2;	

Q.5 CO5	**(a)**	Which keywords are used to handle exceptions in Java? Demonstrate the use of these keywords with the help of small piece of codes.	**6**
	(b)	It is required to have total two threads, both capable of acting as a produce as well as a consumer. If first thread acts as a producer then, the second thread becomes the consumer and vice-versa. They communicate with each other through a buffer, storing one integer number. One of the threads initiates the communication by sending 1 to the other thread. The second thread, on receiving 1 sends 2 to the first thread. On receiving 2, the first thread sends three integer numbers, one by one to the second thread. The second thread consumes the numbers by displaying them. Both threads terminate after that. Note that both threads must be capable of initiating the communication. Write complete multi-threaded program in Java to meet above requirements.	**6**
	(c)	Write a complete multi-threaded program in Java to meet following requirements: - Read matrix [A] m x n - Create m number of threads - Each thread computes summation of elements of one row, i.e. i th row of the matrix is processed by i th thread. Where 0 <= i < m. - Print the results.	**6**